

F-16 High-Fidelity 6DOF Flight Dynamics Simulation Framework

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1 List of Abbreviations

- V : Airspeed (m/s)
- α : Angle of attack (rad)
- β : Sideslip angle (rad)
- γ : Flight-path angle (rad)
- μ : Bank angle (rad)
- p, q, r : Body angular rates (rad/s)
- ϕ, θ, ψ : Euler angles (rad)
- χ : Heading angle (rad)
- u, v, w : Body-axis velocities (m/s)
- x_E, y_E, z_E : Earth-fixed NED position (m)
- q_0, q_1, q_2, q_3 : Quaternion components
- D, L, Y : Drag, lift, side forces (N)
- L_a, M_a, N_a : Aerodynamic moments (N·m)
- T_x, T_y, T_z : Thrust force components (N).
- L_T, M_T, N_T : Thrust-induced moments (N·m).
- $u_{turb}, v_{turb}, w_{turb}$: Dryden turbulence velocity components (m/s).
- $p_{turb}, q_{turb}, r_{turb}$: Angular turbulence velocity components (rad/s).
- $\sigma_u, \sigma_v, \sigma_w$: Turbulence intensity parameters (m/s).
- L_u, L_v, L_w : Turbulence length scales (m).
- m : Total aircraft mass (kg)
- m_{dry} : Dry (fuel-empty) aircraft mass (kg)
- $m_{fuel,0}$: Initial fuel mass (kg)
- $\dot{m}_{fuel}$: Fuel burn rate (kg/s)
- TSFC : Thrust-specific fuel consumption (kg/(N·s))
- τ_{eff} : Effective throttle position, $\tau_{eff} \in [0, 1]$
- τ_{eng} : Engine time constant (s)

- g : Gravity (9.81 m/s^2)
- $I_{xx}, I_{yy}, I_{zz}, I_{xz}$: Inertia tensor elements ($\text{kg}\cdot\text{m}^2$)
- IMU: Inertial Measurement Unit
- $\text{bias}(t)$: bias process
- $\text{noise}(t)$: measurement noise
- $H(s)$: Sensor Transfer Function
- LLA : Longitude, Latitude and altitude

2 Abstract

This project presents a comprehensive high-fidelity open-loop simulation framework for the F-16 Fighter jet, integrating six degrees of freedom flight dynamics. The simulation captures complex nonlinear flight behaviour through a 22-state dynamic model while incorporating realistic sensor measurements. The sensor models provide realistic measurement data incorporating systematic errors and stochastic noise. The framework features translational and rotational dynamics, quaternion-based attitude representation, aerodynamic force and moment modelling, actuator dynamics with physical constraints, a physics-based engine model with rate-limited dynamics, thrust vector control (TVC) effects, a time-varying mass-inertia model, and atmospheric turbulence modelling using the Dryden turbulence model. The simulation environment, developed in MATLAB/Simulink, employs fourth-order Runge-Kutta (RK4) numerical integration. The framework enables a comprehensive evaluation of flight control systems under realistic operational conditions.

3 Introduction

The F-16 stands as one of the most versatile and widely deployed fighter aircraft in modern aviation history. Its exceptional manoeuvrability, coupled with sophisticated flight control systems, has established it as a benchmark for high-performance military aircraft design. Understanding and accurately modelling the complex flight dynamics of such aircraft, along with their associated systems, is paramount for advancing aerospace engineering, developing next-generation control systems, and ensuring mission success in demanding operational environments.

Modern fighter aircraft rely heavily on integrated sensor systems to provide accurate state estimation for flight control, navigation, and mission systems. These sensors, including inertial measurement units (IMU), GNSS receivers, barometric altimeter and others must operate reliably across diverse flight conditions while maintaining precision in the presence of various error sources.

Flight dynamics simulation of high-performance aircraft presents unique challenges due to the inherently nonlinear, coupled, and multivariable nature of aerodynamic phenomena,

particularly at extreme flight conditions. The F-16's capability to operate across a wide flight envelope, including high angles of attack and aggressive manoeuvring scenarios, necessitates sophisticated mathematical models that capture the intricate interplay between aerodynamic forces, inertial effects, control system dynamics, sensor measurements and actuator dynamics.

This report presents a comprehensive open-loop simulation framework for the F-16 aircraft, employing a high-fidelity nonlinear mathematical model that encompasses complete six-degree-of-freedom dynamics. The simulation architecture integrates translational and rotational motion equations, quaternion-based attitude representation to avoid singularities, detailed actuator dynamics with realistic physical constraints, advanced thrust vector control capabilities, comprehensive atmospheric turbulence modelling, a physics-based engine model with asymmetric rate saturation, a time-varying mass model, and sophisticated sensor models incorporating both systematic bias and stochastic noise components.

The mathematical foundation is based on the first principles of Newtonian mechanics, incorporating the Newton-Euler equations of motion along with appropriate coordinate transformations between principal reference frames. The sensor models follow established s-domain transfer function representations with a dual-noise architecture that captures both low-frequency bias drift and high-frequency measurement noise characteristic of real sensor systems.

The simulation environment, developed within MATLAB's Simulink framework, provides a robust computational platform for numerical integration using different Numerical Solvers to solve a set of ordinary differential equations (ODEs), including the Euler Forward and fourth-order Runge-Kutta schemes. In this simulation, the 4th-order Runge-Kutta solver has been chosen as it is more accurate than the Euler forward solver, but at a higher computational cost.

The model architecture is designed with modularity in mind, facilitating future extensions for closed-loop flight control system development and validation. Instead of using built-in Simulink blocks to model the 6-DOF nonlinear dynamics, a user-defined function (MATLAB Function Block) was used to provide design freedom for implementing various design concepts. Figure 1 illustrates the input-output structure of the developed simulation framework, demonstrating the interconnected nature of the various subsystems.

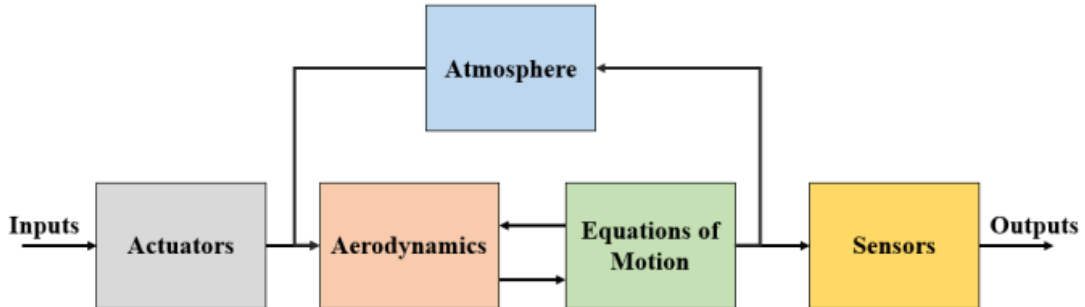


Figure 1: Fighter Aircraft Simulation Model Architecture

3.1 Equations of Motion Block

The equations of motion governing aircraft dynamics are formulated through the application of Newton-Euler principles, which synthesise Newton's second law of motion with coordinate transformation matrices operating between principal reference frames. the math from []

3.2 Translational Dynamics

$$\dot{V} = \frac{Y \sin \beta - mg \sin \gamma - D}{m} + \frac{T_x \cos \beta \cos \alpha + T_y \sin \beta + T_z \sin \beta \cos \alpha}{m} \quad (1)$$

$$\dot{\alpha} = q - (r \sin \alpha + p \cos \alpha) \tan \beta + \frac{mg \cos \gamma \cos \mu - L}{mV \cos \beta + 0.1} + \frac{-T_x \sin \alpha + T_z \cos \alpha}{mV \cos \beta} \quad (2)$$

$$\dot{\beta} = -r \cos \alpha + p \sin \alpha + \frac{-T_x \sin \beta \cos \alpha + T_y \cos \beta - T_z \sin \beta \sin \alpha}{mV} + \frac{Y \cos \beta + mg \cos \gamma \sin \mu}{mV} \quad (3)$$

$$\begin{aligned} \dot{\gamma} = & \frac{-Y \cos \beta \sin \mu + L \cos \mu}{mV} - \frac{T_y}{mV} \cos \beta + \frac{T_x}{mV} (\cos \alpha \sin \beta \sin \mu + \sin \alpha \cos \mu) \\ & + \frac{T_z}{mV} (\sin \alpha \sin \beta \sin \mu - \cos \alpha \cos \mu) \end{aligned} \quad (4)$$

$$\begin{aligned} \dot{\mu} = & \frac{p \cos \alpha + r \sin \alpha}{\cos \beta} + \frac{L}{mV} (\tan \beta + \sin \mu \tan \gamma) + \frac{Y + T_y}{mV} \cos \beta \cos \mu \tan \gamma \\ & - \frac{mg}{V} \tan \beta \cos \mu \cos \gamma + \frac{T_x \sin \alpha - T_z \cos \alpha}{mV} (\tan \beta + \sin \mu \tan \gamma) \\ & - \frac{T_x \cos \alpha + T_z \sin \alpha}{mV} \sin \beta \cos \mu \tan \gamma \end{aligned} \quad (5)$$

$$\begin{aligned} \dot{\chi} = & \frac{Y \cos \beta \cos \mu + L \sin \mu}{mV \cos \gamma} + \frac{T_x}{mV \cos \gamma} (\sin \alpha \sin \mu - \cos \alpha \sin \beta \cos \mu) \\ & + \frac{T_y}{mV \cos \gamma} \cos \mu \cos \beta - \frac{T_z}{mV \cos \gamma} (\cos \mu \sin \alpha \sin \beta + \cos \alpha \sin \mu) \end{aligned} \quad (6)$$

3.3 Rotational Dynamics

$$\dot{p} = \frac{I_{zz}(L_a + L_T) + I_{xz}(N_a + N_T)}{\Delta} + \frac{((I_{xx} - I_{yy})I_{xz} + I_{xz}I_{zz})pq}{\Delta} + \frac{(I_{yy}I_{zz} - I_{zz}^2 - I_{xz}^2)qr}{\Delta} \quad (7)$$

$$\dot{q} = \frac{M_a + M_T}{I_{yy}} - \frac{(I_{xx} - I_{zz})pr}{I_{yy}} - \frac{I_{xz}(p^2 - r^2)}{I_{yy}} \quad (8)$$

$$\dot{r} = \frac{I_{xz}(L_a + L_T) + I_{xx}(N_a + N_T)}{\Delta} + \frac{(I_{xx}^2 - I_{xx}I_{yy} + I_{xz}^2)pq}{\Delta} - \frac{I_{xz}(I_{xx} - I_{yy} + I_{zz})qr}{\Delta} \quad (9)$$

where $\Delta = I_{xx} \cdot I_{zz} - I_{xz}^2$ and I_{xx}, I_{yy}, I_{zz} are the moments of inertia.

3.4 Quaternion Kinematics

$$\dot{\phi} = p + q \sin \phi \tan \theta + r \cos \phi \tan \theta \quad (10)$$

$$\dot{\theta} = q \cos \phi - r \sin \phi \quad (11)$$

$$\dot{\psi} = \frac{q \sin \phi + r \cos \phi}{\cos \theta} \quad (12)$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes [0, p, q, r]^\top \quad (13)$$

3.5 NED Position Rates

$$\dot{x}_E = V \cos \chi \cos \gamma, \quad \dot{y}_E = V \sin \chi \cos \gamma, \quad \dot{z}_E = -V \sin \gamma \quad (14)$$

3.6 Body-Fixed Velocities

$$\dot{u} = rv - qw + g \sin \theta + \frac{T_x - D \cos \alpha \cos \beta + L \sin \alpha}{m} \quad (15)$$

$$\dot{v} = -ru + pw - g \cos \theta \sin \phi + \frac{T_y + Y}{m}, \quad (16)$$

$$\dot{w} = qu - pv - g \cos \theta \cos \phi + \frac{T_z - D \sin \alpha \cos \beta - L \cos \alpha}{m}. \quad (17)$$

where V is flight speed, m is the mass of the aircraft fighter, and g is the gravity acceleration. α and β represent the angle of attack and sideslip angle, respectively. γ , χ and μ denote flight path angle, velocity heading angle and roll angle around the velocity vector, respectively. p , q and r represent roll, pitch and yaw angular rate in the body coordinate system, respectively. x_E , y_E and z_E are the flight position coordinates.

4 Actuator Models

The actuators are critical components of the flight control system, providing the motive power needed to move the flight control surfaces and manage thrust vectoring. The F-16 flight control system encompasses both conventional aerodynamic control surfaces (aileron, elevator, rudder) and advanced thrust vector control (TVC) capabilities for enhanced manoeuvrability.

4.1 Control Surface Actuators

The primary flight control surfaces, namely aileron (δ_a), elevator (δ_e), and rudder (δ_r), are shown in Figure 3. These conventional control surfaces provide the primary means of aircraft attitude and trajectory control through aerodynamic moment generation.

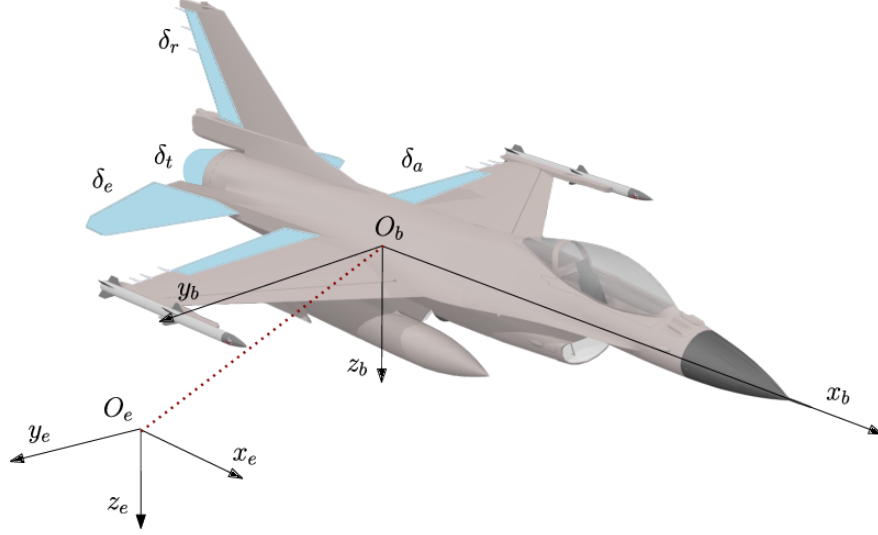


Figure 2: F-16 aerodynamic control surfaces

The actuator mathematical model for each control surface is represented as a second-order transfer function with pure time delay, dead zone and backlash:

$$\delta_{i,\text{act}}(s) = K e^{-s\tau} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \mathcal{B}_{b_i} \mathcal{D}_{d_i} \quad (18)$$

For implementation, the delay term $e^{-s\tau}$ can be approximated using 1-order Padé expansions as:

$$e^{-s\tau} \approx \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s} \quad (19)$$

where:

- $\delta_{i,\text{com}}$: commanded deflection,
- $\delta_{i,\text{act}}$: actual actuator deflection,
- K : steady-state gain,
- ω_n : natural frequency [rad/s],
- ζ : damping ratio,
- τ : time delay,
- $\mathcal{D}_{d_i}$: dead-zone with width d_i ,
- $\mathcal{B}_{b_i}$: backlash with width b_i ,
- $i \in \{a, e, r\}$: aileron, elevator, and rudder, respectively.

4.2 Thrust Vector actuators

In addition to conventional control surfaces, the F-16 model incorporates thrust vector control capabilities through one movable engine nozzle. The TVC system provides two degrees of freedom: pitch vectoring (δ_z) and yaw vectoring (δ_y), enabling direct thrust-based control moments that are particularly effective at high angles of attack where aerodynamic control authority may be reduced.

The nozzle deflection actuators share the same second-order transfer function structure as the aerodynamic control surfaces.

4.3 Engine Thrust Model

The engine thrust magnitude is governed by a nonlinear physics-based model that accounts for engine dynamics, asymmetric rate saturation, density altitude correction, and Mach number effects.

4.3.1 Engine Dynamics

The engine dynamics are modelled using a simple linear first-order lag:

$$\frac{\tau_{\text{eff}}(s)}{\tau_{\text{cmd}}(s)} = \frac{1}{\tau_{\text{eng}}s + 1}, \quad (20)$$

where $\tau_{\text{eng}} = 3$ s is the engine time constant.

In addition to the first-order lag, the engine model includes both position and rate limits. The throttle command is constrained to the interval $[0, 1]$, ensuring physically admissible input values.

Note: the rate saturation limits are set below the natural response rate of the lag, $1/\tau_{\text{eng}} \approx 0.33 \text{ s}^{-1}$. As a result, the rate limiter becomes the active constraint during aggressive throttle transients, while the position saturation prevents the commanded signal from exceeding the admissible range.

4.3.2 Thrust Law

The engine thrust is computed from the effective throttle with altitude and Mach corrections:

$$T_E = T_{idle} + \tau_{eff} (T_{max,SL} - T_{idle}) \left(\frac{\rho}{\rho_0} \right)^{0.9} f_M \quad (21)$$

where:

- $T_{max,SL}$ — is the maximum sea-level static thrust,
- T_{idle} — idle thrust,
- ρ_0 — sea-level air density,
- $(\rho/\rho_0)^{0.9}$ — sub-linear density correction (ram and mass-flow effect),
- $f_M = \max(0.60, 1 - 0.10 M)$ — Mach correction factor.

4.4 Time-Varying Mass, Inertia and CG Model

Aircraft mass varies continuously during flight due to fuel consumption. The mass model captures this effect through a thrust-proportional burn rate governed by the thrust-specific fuel consumption (TSFC):

$$\dot{m}(t) = -\frac{1}{I_{\text{sp}} g_0} T_{\text{ref}}(t) \quad (22)$$

where I_{sp} is the specific impulse of the engine and $g_0 \approx 9.81 \text{ m/s}^2$ is the gravitational acceleration at the Earth's surface.

The total aircraft mass is integrated at each time step with a hard lower bound at the dry mass:

$$m[k+1] = \max(m_{\text{dry}}, m[k] + \Delta t \dot{m}[k]) \quad (23)$$

where m_{dry} is the fuel-empty structural mass, and the guard prevents the integrator from crossing below the physical lower limit once all fuel is exhausted.

The inertia tensor is assumed to scale linearly with total mass (central tank assumption), which is consistent with a fuel tank located near the centre of gravity:

$$I_{xx}(t) = I_{xx,0} \frac{m(t)}{m_0}, \quad I_{yy}(t) = I_{yy,0} \frac{m(t)}{m_0}, \quad I_{zz}(t) = I_{zz,0} \frac{m(t)}{m_0}, \quad I_{xz}(t) = I_{xz,0} \frac{m(t)}{m_0} \quad (24)$$

where $m_0 = m_{\text{dry}} + m_{\text{fuel},0}$ is the initial total mass, the subscript 0 denotes initial values, and m_{dry} is the fuel-empty structural mass.

As fuel is consumed during flight, the total aircraft mass decreases according to the fuel-burning model described previously. Since the fuel is stored at a fixed location relative to the airframe, the centre of gravity (CG) migrates longitudinally as the fuel fraction changes. The instantaneous CG position along the body x -axis is computed as the mass-weighted average of the empty airframe contribution and the remaining fuel contribution:

$$x_{\text{cg}}(t) = \frac{m_{\text{dry}} x_{\text{cg},e} + m_f(t) x_{\text{cg},f}}{m(t)} \quad (25)$$

where $x_{\text{cg},e}$ is the empty-mass CG station, $x_{\text{cg},f}$ is the fuel-tank CG station, $m_f(t)$ is the instantaneous fuel mass, and $m(t)$ is the total instantaneous mass.

The longitudinal CG shift $\Delta x_{\text{cg}}(t) = x_{\text{cg}}(t) - x_{\text{cg},0}$ directly modifies the aerodynamic pitching moment. These time-varying mass, CG and inertia values are passed explicitly into the equations of motion at each RK4 sub-step, ensuring that the integration remains consistent with the instantaneous aircraft state.

4.5 Control Input Constraints

All actuators operate within physical limitations that must be enforced to ensure realistic simulation behaviour. The control input vector is defined as:

$$\mathbf{U} = [\delta_a, \delta_e, \delta_r, \delta_y, \delta_z, T_E]^T \quad (26)$$

The operational limits and system parameters for each control input are summarised in Table 1.

Table 1: Actuator Model Parameters

Control Surface	Position Limit	Rate Limit	ω_n (rad/s)	ζ	τ (ms)	
$\delta_{a,e,r}$	± 35	$\pm 80/\text{s}$	75	0.6	8	
$\delta_{y,z}$	± 20	$\pm 60/\text{s}$	75	0.7	5	
T_E	2000–90000 kN	$+0.20/\text{s}, -0.25/\text{s}$	—	—	—	

These saturation limits are enforced in the simulation to prevent unrealistic control commands and ensure the actuator responses remain within their physical capabilities. The combination of conventional control surfaces and thrust vectoring provides the F-16 with exceptional control authority across its entire flight envelope.

Figures 4 and 5 illustrate the Simulink implementation of the Thrust vector and engine actuation system.

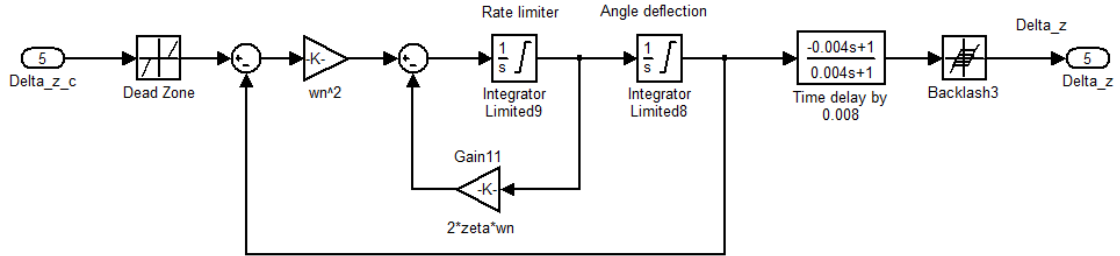


Figure 3: Simulink block implementation of the Thrust vector Actuation System

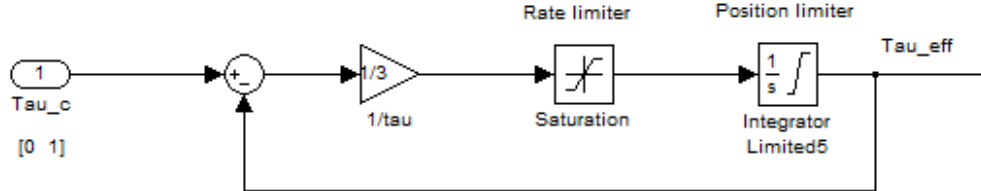


Figure 4: Simulink block implementation of the engine Actuation System

This implementation of the actuator models provides a realistic representation of modern flight control actuator systems, including the effects of digital processing delays that are characteristic of contemporary fly-by-wire flight control architectures. The actuators parameters can be adjusted based on specific hardware configurations and system requirements.

5 Atmospheric Model (ISA)

As an aircraft operates at various altitudes, the aerodynamic coefficients need to account for changes in density as represented by a standard atmosphere model. The International Standard Atmosphere (ISA) model used in this simulation provides standard atmospheric values for air temperature T , pressure P , density ρ , speed of sound a , and dynamic pressure q_{bar} , as a function of the altitude h and true airspeed V_T .

5.1 Reference Constants

The ISA model uses the following sea-level reference conditions:

$$T_0 = 288.15 \text{ K} \quad (\text{sea-level temperature}) \quad (27)$$

$$P_0 = 101325 \text{ Pa} \quad (\text{sea-level pressure}) \quad (28)$$

$$R = 287.05 \text{ J/(kg}\cdot\text{K)} \quad (\text{specific gas constant for air}) \quad (29)$$

$$g = 9.80665 \text{ m/s}^2 \quad (\text{standard gravity}) \quad (30)$$

$$\gamma = 1.4 \quad (\text{heat capacity ratio}) \quad (31)$$

5.2 Temperature and Pressure Calculation

The atmospheric properties are computed using the hydrostatic equation and the ideal gas law based on altitude layers.

Troposphere ($h \leq h_1 = 11000 \text{ m}$):

$$T = T_0 + L_1 h \quad (32)$$

$$P = P_0 \left(\frac{T}{T_0} \right)^{-\frac{g}{RL_1}} \quad (33)$$

where $L_1 = -0.0065 \text{ K/m}$ is the temperature lapse rate in the troposphere.

Above Troposphere ($h > 11000 \text{ m}$):

For altitudes above 11 km, an isothermal approximation is used:

$$T_1 = T_0 + L_1 h_1 \quad (34)$$

$$P_1 = P_0 \left(\frac{T_1}{T_0} \right)^{-\frac{g}{RL_1}} \quad (35)$$

$$T = T_1 \quad (36)$$

$$P = P_1 \exp \left(-\frac{g(h - h_1)}{RT_1} \right) \quad (37)$$

5.3 Derived Quantities

Once temperature and pressure are determined, the remaining atmospheric properties are calculated using fundamental thermodynamic relationships:

Air density:

$$\rho = \frac{P}{RT} \quad (38)$$

Dynamic pressure:

$$q_{\text{bar}} = \frac{1}{2} \rho V_T^2 \quad (39)$$

Speed of sound:

$$a = \sqrt{\gamma RT} \quad (40)$$

Mach number:

$$M = \frac{V_T}{a} \quad (41)$$

5.4 WGS-84 gravity model

The WGS-84 gravity model considers the variance in gravitational acceleration based on altitude and latitude, attributable to the Earth's ellipsoidal form. The standard gravitational acceleration at sea level is adjusted using the subsequent empirical formula:

$$g_0(\phi) = 9.780327 (1 + 0.0053024 \sin^2 \phi - 0.0000058 \sin^2(2\phi)) \quad (42)$$

The gravitational force at a specific geodetic latitude (ϕ) and elevation (h) is calculated as:

$$g(h, \phi) = g_0(\phi) \left(\frac{R_E}{R_E + h} \right)^2 \quad (43)$$

where ($R_E \approx 6371000$ m) represents the average radius of the Earth. This model facilitates precise depiction of the aircraft's weight force and potential energy fluctuations at varying altitudes.

6 Atmospheric Turbulence Modeling

Real-world flight operations are significantly influenced by atmospheric turbulence, which originates from atmospheric pressure instabilities and non-uniform temperature distributions, introducing stochastic disturbances that affect aircraft stability, handling qualities, and structural loads. The incorporation of accurate turbulence models is essential for comprehensive flight simulation and robust control system validation under realistic atmospheric conditions.

The inherently stochastic nature of atmospheric turbulence cannot be modelled exactly, necessitating statistical approaches that capture the essential spectral characteristics of turbulent velocity fields. Several well-established turbulence models have been used in the literature to implement turbulence, among which the Dryden and von Kármán spectral models

have achieved widespread acceptance due to their physical realism, mathematical simplicity, and computational efficiency.

6.1 Dryden Turbulence Model

The Dryden turbulence model is widely adopted in aerospace engineering due to its mathematical tractability and good correlation with measured atmospheric turbulence spectra. The model generates correlated turbulence velocity components through shaping filters that transform white noise inputs into realistic atmospheric disturbances with appropriate spectral characteristics.

The Dryden model defines three turbulence velocity components corresponding to longitudinal (u_{turb}), lateral (v_{turb}), and vertical (w_{turb}) atmospheric disturbances. These components are generated using the following continuous-domain transfer functions:

$$G_{ug}(s) = \sigma_u \sqrt{\frac{2L_u}{\pi V}} \cdot \frac{1}{1 + \frac{L_u}{V}s} \quad (44)$$

$$G_{vg}(s) = \sigma_v \sqrt{\frac{2L_v}{\pi V}} \cdot \frac{1 + \frac{2\sqrt{3}L_v}{V}s}{\left(1 + \frac{2L_v}{V}s\right)^2} \quad (45)$$

$$G_{wg}(s) = \sigma_w \sqrt{\frac{2L_w}{\pi V}} \cdot \frac{1 + \frac{2\sqrt{3}L_w}{V}s}{\left(1 + \frac{2L_w}{V}s\right)^2} \quad (46)$$

where σ_u , σ_v , σ_w are the turbulence intensity parameters, L_u , L_v , L_w are the turbulence length scales, and V is the aircraft airspeed. The longitudinal turbulence exhibits first-order dynamics, while the lateral and vertical components follow second-order characteristics to capture the more complex cross-coupling effects observed in atmospheric measurements.

6.2 Discrete-Time Implementation

The continuous-domain Dryden filters are converted to discrete-time equivalents using the bilinear transformation (Tustin method) to enable numerical integration within the simulation framework. This transformation preserves the essential spectral characteristics while ensuring numerical stability at the chosen integration timestep.

The bilinear transformation $s = \frac{2}{dt} \frac{z-1}{z+1}$ is applied to each transfer function, resulting in discrete-time difference equations suitable for real-time implementation. The transformed filters maintain appropriate filter memory states to ensure continuity and realistic temporal correlation in the generated turbulence sequences.

6.3 Turbulence-Induced Aerodynamic Perturbations

Atmospheric turbulence modifies the relative wind velocity vector experienced by the aircraft. The turbulence components in the body-fixed reference frame are denoted as u_{turb} , v_{turb} , and w_{turb} , representing longitudinal, lateral, and vertical wind disturbances, respectively.

These components are generated through spectral filtering of white noise to produce realistic atmospheric disturbances with appropriate temporal and spatial correlation characteristics.

6.3.1 Linear Velocity Perturbations

The relative wind velocity experienced by the aircraft is computed by subtracting the turbulence velocity components from the aircraft's body-frame velocity components:

$$u_{rel} = u - u_{turb} \quad (47)$$

$$v_{rel} = v - v_{turb} \quad (48)$$

$$w_{rel} = w - w_{turb} \quad (49)$$

The magnitude of the relative velocity and the aerodynamic angles are computed from the perturbed velocity components:

$$V_{rel} = \sqrt{u_{rel}^2 + v_{rel}^2 + w_{rel}^2} \quad (50)$$

$$\alpha = \arctan \left(\frac{w_{rel}}{u_{rel}} \right) \quad (51)$$

$$\beta = \arcsin \left(\frac{v_{rel}}{V_{rel}} \right) \quad (52)$$

where α is the angle of attack and β is the sideslip angle, both modified by the presence of atmospheric turbulence.

6.3.2 Angular Rate Perturbations

In addition to linear velocity turbulence, atmospheric disturbances induce rotational perturbations on the aircraft. These angular rate disturbances arise from spatial gradients in the turbulent velocity field and can be approximated using the following relations based on MIL-STD-1797A:

$$p_{gust} = \frac{\pi}{4b} (w_{turb} - v_{turb}) \quad (53)$$

$$q_{gust} = \frac{\pi}{3b} w_{turb} \quad (54)$$

$$r_{gust} = \frac{\pi}{4b} (v_{turb} + w_{turb}) \quad (55)$$

where b is the aircraft wingspan and p_{gust} , q_{gust} , r_{gust} represent roll, pitch, and yaw rate disturbances, respectively. These angular perturbations modify the effective body angular rates experienced by the aerodynamic surfaces, thereby affecting the damping and rate-dependent moment contributions.

The relative angular rates become:

$$p_{rel} = p - p_{gust} \quad (56)$$

$$q_{rel} = q - q_{gust} \quad (57)$$

$$r_{rel} = r - r_{gust} \quad (58)$$

7 Aerodynamic Force and Moment

7.1 Aerodynamic Force

The aerodynamic forces are computed using the disturbed values V_{rel} , α , and β in place of the undisturbed values.

The aerodynamic forces are initially computed in the wind-axis, where:

$$L = q_{bar} S C_L \quad (\text{Lift, perpendicular to } V_{rel}) \quad (59)$$

$$D = q_{bar} S C_D \quad (\text{Drag, parallel to } V_{rel}) \quad (60)$$

$$Y = q_{bar} S C_Y \quad (\text{Side force}) \quad (61)$$

where S is the reference wing area, and q_{bar} is the dynamic pressure calculated using the relative velocity:

$$q_{bar} = \frac{1}{2} \rho V_{rel}^2 \quad (62)$$

C_L, C_D, C_Y are the aerodynamic force coefficients which are evaluated using perturbed aerodynamic angles.

For a typical high-performance fighter aircraft, such as the F-16, can be represented as:

$$C_L = C_{L_0} + C_{L_\alpha} \alpha + C_{L_q} \frac{q_{rel} \bar{c}}{2V_{rel}} + C_{L_{\delta_e}} \delta_e \quad (63)$$

$$C_D = C_{D_0} + C_{D_\alpha} \alpha^2 + C_{D_{\delta_e}} |\delta_e| \quad (64)$$

$$C_Y = C_{Y_\beta} \beta + C_{Y_p} \frac{p_{rel} b}{2V_{rel}} + C_{Y_r} \frac{r_{rel} b}{2V_{rel}} + C_{Y_{\delta_a}} \delta_a + C_{Y_{\delta_r}} \delta_r \quad (65)$$

7.2 Aerodynamic Moment

The aerodynamic moments about the aircraft center of gravity are similarly computed using the relative velocity and angular rates.

The aerodynamic moments about the body-fixed axes are:

$$L_{aero} = q_{bar} S b C_\ell \quad (66)$$

$$M_{aero} = q_{bar} S \bar{c} C_m \quad (67)$$

$$N_{aero} = q_{bar} S b C_n \quad (68)$$

where b is the wingspan and $\bar{c}$ is the mean aerodynamic chord.

C_ℓ, C_m, C_n are the moment coefficients represented as:

$$C_\ell = C_{\ell_\beta} \beta + C_{\ell_p} \frac{p_{rel} b}{2V_{rel}} + C_{\ell_r} \frac{r_{rel} b}{2V_{rel}} + C_{\ell_{\delta_a}} \delta_a + C_{\ell_{\delta_r}} \delta_r \quad (69)$$

$$C_m = C_{m_0} + C_{m_\alpha} \alpha + C_{m_q} \frac{q_{rel} \bar{c}}{2V_{rel}} + C_{m_{\delta_e}} \delta_e \quad (70)$$

$$C_n = C_{n_\beta} \beta + C_{n_p} \frac{p_{rel} b}{2V_{rel}} + C_{n_r} \frac{r_{rel} b}{2V_{rel}} + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r \quad (71)$$

Table 2: Aerodynamic Stability Derivatives

Derivative	Value	Derivative	Value
C_{L_α}	5.8 rad ⁻¹	C_{ℓ_β}	-0.12 rad ⁻¹
C_{D_α}	0.15 -	C_{ℓ_p}	-0.40 -
C_{Y_β}	-0.75 rad ⁻¹	C_{ℓ_r}	0.15 -
C_{L_q}	8.0 -	C_{m_α}	-1.2 rad ⁻¹
C_{Y_p}	0.05 -	C_{m_q}	-12.0 -
C_{Y_r}	0.30 -	C_{n_β}	0.10 rad ⁻¹
$C_{L_{\delta_e}}$	0.40 rad ⁻¹	C_{n_p}	-0.05 -
$C_{Y_{\delta_r}}$	0.15 rad ⁻¹	C_{n_r}	-0.20 -
$C_{\ell_{\delta_a}}$	0.15 rad ⁻¹	$C_{m_{\delta_e}}$	-1.5 rad ⁻¹
$C_{\ell_{\delta_r}}$	0.01 rad ⁻¹	$C_{n_{\delta_a}}$	0.01 rad ⁻¹
-	-	$C_{n_{\delta_r}}$	-0.12 rad ⁻¹

8 Thrust Vectoring

Modern fighter aircraft like the F-16 employ thrust vectoring systems to enhance manoeuvrability and control authority. The thrust vectoring system redirects the engine exhaust flow through movable nozzles, creating additional control forces and moments which provide additional control authority, particularly beneficial during high-angle-of-attack flight or when aerodynamic control surfaces become less effective.

8.1 Force Components

The thrust vector components in the body frame are calculated based on the nozzle deflection angles. The thrust force decomposed into body-axis components according to the nozzle orientation:

$$T_x = \zeta_f T_E \cos(\delta_z) \cos(\delta_y) \quad (72)$$

$$T_y = \zeta_f T_E \cos(\delta_z) \sin(\delta_y) \quad (73)$$

$$T_z = \zeta_f T_E \sin(\delta_z) \quad (74)$$

where δ_y and δ_z are the nozzle deflection angles in the yaw and pitch planes, respectively, T_E is the engine thrust magnitude, and ζ_f is the combined nozzle loss coefficient accounting for efficiency losses in the vectoring system.

8.2 Induced Moments

The thrust forces applied at the engine location create additional moments about the aircraft's centre of gravity. These thrust-induced moments are calculated using the cross product of the moment arm and thrust force vector:

$$L_T = y_E T_z - z_E T_y \quad (75)$$

$$M_T = z_E T_x - x_E T_z \quad (76)$$

$$N_T = x_E T_y - y_E T_x \quad (77)$$

where (x_E, y_E, z_E) represents the engine location relative to the centre of gravity in body coordinates.

9 Sensors Model

Modern fighter aircraft are critically dependent on integrated sensor systems to provide accurate state estimation for flight control, navigation, and mission systems. The F-16 simulation framework incorporates a comprehensive multisensor model that includes inertial measurement units (IMU) with realistic error characteristics and dynamic responses.

10 Sensor Model Architecture

The multi-sensor suite is represented by a unified, five-stage processing chain that captures the principal characteristics of practical sensing systems:

1. **Dynamic filtering:** sensor dynamics described by a transfer function.
2. **Scale factor:** sensor's sensitivity gain.
3. **Bias injection:** addition of slowly varying systematic offsets.
4. **Noise corruption:** addition of measurement noise (high-frequency).
5. **Time delay:** latency due to digital processing and communication.

A general s-domain model for a single measurement channel is:

$$Y(s) = K U(s) H(s) D(s) + B(s) + N(s) \quad (78)$$

where $\mathcal{H}$ represents the sensor transfer function, K is the scale factor, $U(s)$ the true input, T the pure time delay, $B(s)$ the bias process, $D(s)$ is the par   delay operator and $N(s)$ the additive measurement noise.

10.1 Bias dynamics

Sensor bias denotes a slowly varying systematic error caused by temperature, ageing, calibration drift and similar effects. A common and practical model is the first-order Gauss-Markov process

$$\dot{b}(t) = -\frac{1}{\tau} b(t) + w(t) \quad (79)$$

where $\tau > 0$ is the bias correlation time and $w(t)$ is zero-mean white driving noise with variance σ_w^2 . The corresponding transfer function from the driving noise to the bias is

$$\frac{B(s)}{W(s)} = \frac{1}{\tau s + 1}. \quad (80)$$

where the bias variance is

$$\sigma_b^2 = \frac{\sigma_w^2 \tau}{2} \implies \sigma_w^2 = \frac{2\sigma_b^2}{\tau} \quad (81)$$

10.2 Measurement noise

High-frequency measurement disturbances (e.g. ADC quantisation, electronic noise, small-scale turbulence) are modelled as additive, zero-mean Gaussian white noise:

$$n(t) \sim \mathcal{N}(0, \sigma_n^2), \quad (82)$$

where σ_n^2 denotes the noise variance.

10.3 Inertial Measurement Unit (IMU)

The IMU provides three-axis acceleration and angular rate measurements through accelerometer and gyroscope sensor clusters. These sensors are fundamental to the aircraft's inertial navigation system and flight control loops. The sensor families follow the general architecture above.

10.3.1 Accelerometer

The accelerometer measurement vector is modelled as

$$\mathbf{Y}_{\text{acc}}(s) = K_{\text{acc}} \mathbf{F}_{\text{true}}(s) H_{\text{acc}}(s) e^{-sT} + \mathbf{B}_{\text{acc}}(s) + \mathbf{N}_{\text{acc}}(s) \quad (83)$$

with a second-order dynamics approximation

$$H_{\text{acc}}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad (84)$$

Parameter values are:

- natural frequency: $\omega_n = 100 \text{ rad/s}$ ($\approx 15.9 \text{ Hz}$)

- damping ratio: $\zeta = 0.7$ (underdamped)
- scale factor: $K_{\text{acc}} = 1.0 \pm 0.005$ (0.5% error)
- bias correlation time: $\tau_{\text{acc}} = 300$ s
- bias std: $\sigma_{b,\text{acc}} = 0.01$ m/s²
- noise variance: $\sigma_{n,\text{acc}}^2 = 0.001$ m/s²/√Hz.

10.3.2 Gyroscope

The gyroscope measurement is modelled as

$$\mathbf{\Omega}_{\text{meas}}(s) = K_{\text{gyro}} \mathbf{\Omega}_{\text{true}}(s) H_{\text{gyro}}(s) e^{-sT} + \mathbf{B}_{\text{gyro}}(s) + \mathbf{N}_{\text{gyro}}(s) \quad (85)$$

with a first-order dynamics approximation

$$H_{\text{gyro}}(s) = \frac{1}{\tau_{\text{gyro,d}} s + 1} \quad (86)$$

Parameters are:

- dynamics time constant: $\tau_{\text{gyro,d}} = 0.01$ s (≈ 100 Hz bandwidth)
- scale factor: $K_{\text{gyro}} = 1.0 \pm 0.002$ (0.2% error)
- bias correlation time: $\tau_{\text{gyro,b}} = 300$ s
- bias std: $\sigma_{b,\text{gyro}} = 0.001$ rad/s
- noise variance: $\sigma_{n,\text{gyro}}^2 = 1 \times 10^{-4}$ rad/s/√Hz.

10.4 Air Data System (ADS)

The air data system provides aerodynamic state information, including airspeed and flow angles. These measurements are modelled with second-order dynamics, transport delays, and stochastic errors.

10.4.1 Airspeed Sensor

The indicated airspeed measurement is modelled as:

$$v_{\text{meas}}(s) = K_v v_{\text{true}}(s) H_v(s) e^{-sT} + \mathbf{B}_v(s) + \mathbf{N}_v(s) \quad (87)$$

with second-order dynamics:

$$H_{V_a}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (88)$$

Parameter values:

- natural frequency: $\omega_n = 20$ rad/s
- damping ratio: $\zeta = 0.7$
- transport delay: $\tau_p = 0.005$ s
- scale factor: $K_{V_a} = 1.002$
- bias correlation time: $\tau_{gm} = 1000$ s
- bias std: $\sigma_{gm} = 5.0 \times 10^{-2}$ m/s
- noise std: $\sigma_n = 1.0 \times 10^{-1}$ m/s/ $\sqrt{\text{Hz}}$

10.4.2 Flow Angle Sensors (AoA and Sideslip)

The Angle of Attack (α) and Sideslip (β) vanes are modeled using the same second-order architecture :

$$\boldsymbol{\alpha}_{\text{meas}}(s) = K_{\text{alpha}} \boldsymbol{\alpha}_{\text{true}}(s) H_{\text{alpha}}(s) e^{-sT} + \boldsymbol{B}_{\text{alpha}}(s) + \boldsymbol{N}_{\text{alpha}}(s) \quad (89)$$

The dynamics follow the standard form:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (90)$$

Parameter values for α and β vanes:

- natural frequency: $\omega_n = 20$ rad/s
- damping ratio: $\zeta = 0.7$
- transport delay: $\tau_p = 0.005$ s
- scale factors: $K_{\alpha} = 1.001$, $K_{\beta} = 0.999$
- bias correlation time: $\tau_{gm} = 1000$ s
- bias std: $\sigma_{gm} = 5.0 \times 10^{-4}$ rad
- noise std: $\sigma_n = 1.0 \times 10^{-3}$ rad/ $\sqrt{\text{Hz}}$

10.5 Global Positioning System (GPS)

The GPS provides three-axis position and velocity measurements in the LLA frame.

10.5.1 GPS

The GPS measurement vector is modelled as:

$$\mathbf{Y}_{\text{gps}}(s) = \mathbf{F}_{\text{true}}(s) H_{\text{gps}}(s) e^{-sT} + \mathbf{B}_{\text{gps}}(s) + \mathbf{N}_{\text{gps}}(s) \quad (91)$$

with a First-order dynamics approximation:

$$H_{\text{gyro}}(s) = \frac{1}{\tau_{\text{gyro,d}}s + 1} \quad (92)$$

Parameter values are:

- dynamics time constant: $\tau_{\text{gps}} = 0.01 \text{ s}$ ($\approx 100 \text{ Hz}$ bandwidth)
- bias correlation time: $\tau_{\text{acc}} = 300 \text{ s}$
- bias std: $\sigma_{b,\text{acc}} = 0.01 \text{ m/s}^2$
- noise variance: $\sigma_{n,\text{acc}}^2 = 0.001 \text{ m/s}^2/\sqrt{\text{Hz}}$.

Figure 5 illustrates the Simulink implementation of this architecture for the accelerometer, demonstrating the cascaded processing stages from true input to corrupted measurement output.

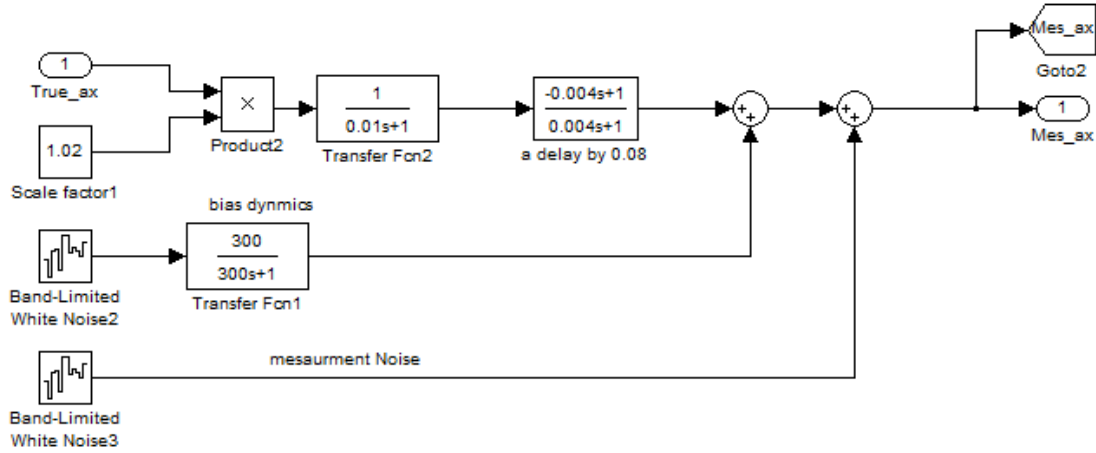


Figure 5: Simulink block implementation of the gyroscope sensor model

11 Simulation Setup

The simulation has been implemented in MATLAB R2011a with the following initial conditions and parameters:

- Integrator: RK4, $\Delta t = 0.001 \text{ s}$, $T = 60 \text{ s}$.
- Initial: $V = 150 \text{ m/s}$, $\alpha = 0.01 \text{ rad}$, $\beta = 0$, $p = q = r = 0$.

- $u = V \cos \alpha$, $v = 0$, $w = V \sin \alpha$.
- quaternion $(1, 0, 0, 0)$, $x_E = 0$, $y_E = 0$, $z_E = -5000$ m (i.e. $h = 5000$ m).
- Aircraft dry mass: $m_{dry} = 8500$ kg; initial fuel: $m_{fuel,0} = 500$ kg; total initial mass: $m_0 = 9000$ kg.
- TSFC $= 2.5 \times 10^{-5}$ kg/(N s).
- Initial inertia: $I_{xx,0} = 15000$, $I_{yy,0} = 25000$, $I_{zz,0} = 35000$, $I_{xz,0} = 1500$ kg m².
- Engine time constant: $\tau_{eng} = 3$ s; rate limits $R = 0.20$ s⁻¹,
- $g = 9.81$ m/s².
- The nozzle loss coefficient $\zeta_f = 1$.
- $S = 30$ m² wing area; $b = 9.5$ m wingspan; $\bar{c} = 3.5$ m mean aerodynamic chord.
- Engine offset: $(-0.5, 0, 0)$ m relative to CG.
- Turbulence: Dryden model with light intensity, persistent filter states cleared before each run.

12 Results

The following results showcase the flight dynamic response.

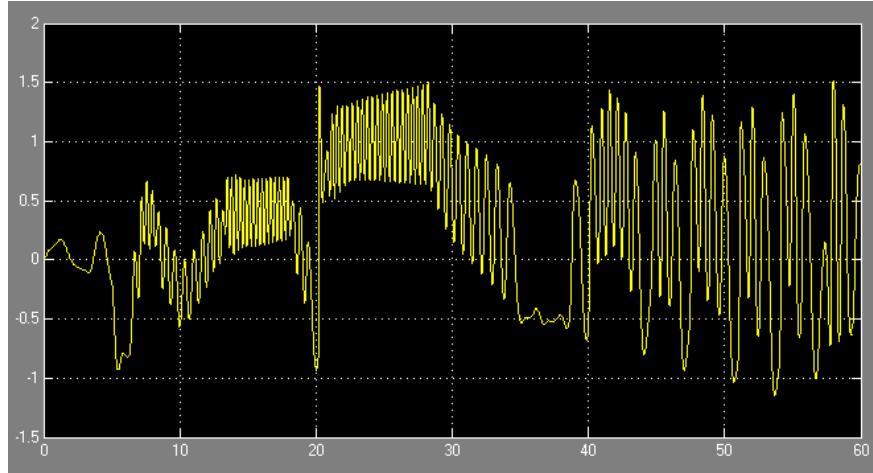


Figure 6: Angle of attack α .

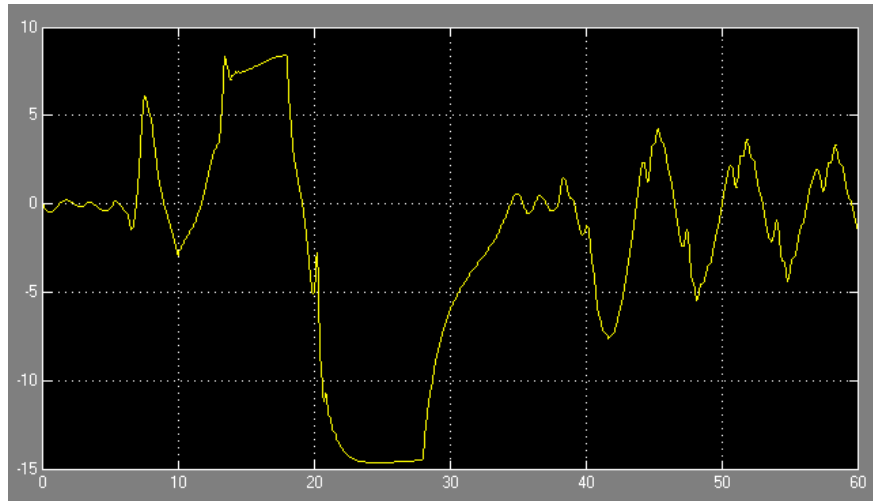


Figure 7: Yaw rate r .

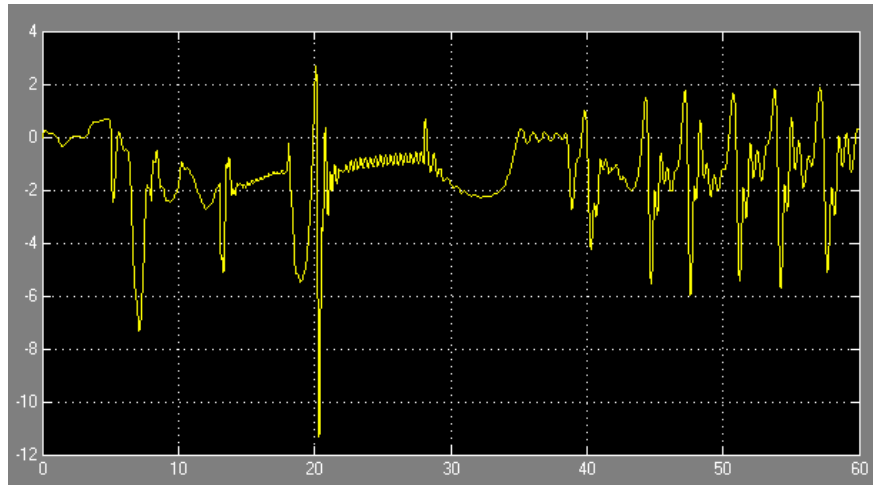


Figure 8: Pitch rate q .

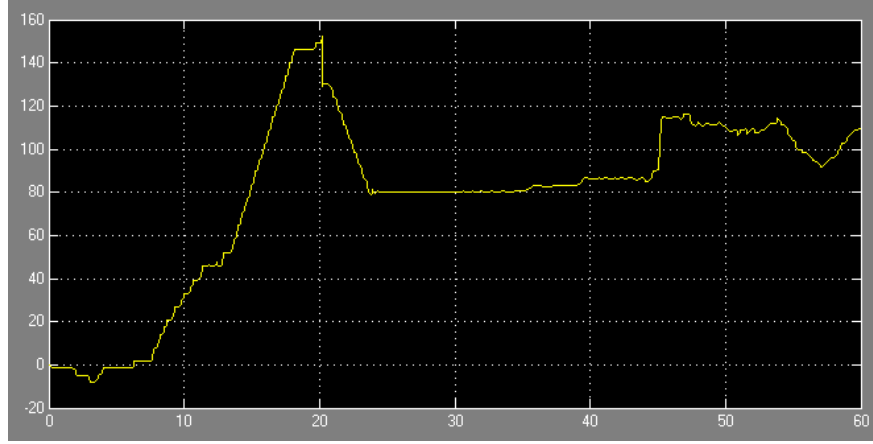


Figure 9: Bank angle μ .

13 Conclusion

This work presents a comprehensive, high-fidelity simulation framework for the F-16 fighter aircraft that successfully integrates six-degree-of-freedom nonlinear flight dynamics with realistic sensor models, a physics-based propulsion model, a time-varying mass, cg and inertia model, and atmospheric disturbances. The implementation demonstrates several key achievements that establish a robust foundation for advanced flight control research.

The numerical architecture, employing fourth-order Runge-Kutta integration with 1 ms timestep, maintained stable behaviour throughout extended simulation periods without numerical instabilities. The adoption of quaternion-based attitude representation effectively eliminated the gimbal lock singularities inherent to Euler angle formulations, ensuring stable flight across the entire attitude envelope, including extreme pitch conditions near ± 90 .

The actuator subsystem accurately captures the physical constraints, dynamic characteristics and the nonlinear elements of modern flight control systems. Position and rate saturation limits, combined with transfer-function dynamics and realistic time delays, Dead-zone and backlash provide realistic representations of control-surface and thrust vectoring actuator behaviour. The observed oscillatory responses in the angular rates (q , r), angle of attack (α) reflect the expected behaviour of the open-loop system in the absence of stabilising feedback control.

The time-varying mass model introduces fuel burn as a continuous state driven by the TSFC-weighted engine thrust. The resulting mass reduction and proportional inertia scaling are propagated into the equations of motion at each integration step, capturing the progressive reduction in rotational resistance as fuel is consumed. This provides a physically consistent foundation for extended-duration mission simulations where mass variation becomes non-negligible.

Integration of atmospheric turbulence through the discrete-time Dryden model successfully introduces realistic disturbances while maintaining numerical stability via bilinear transformation. Turbulence forces and moments provide essential environmental excitation for a complete validation of the control system under operational conditions.

The multi-sensor architecture implements a sophisticated dual-noise error model capturing both systematic bias drift through Gauss-Markov processes and high-frequency measurement noise.

This framework provides a robust testbed for advanced research in flight control design, sensor fusion, enabling rigorous evaluation of control architectures under realistic operational conditions such as high-fidelity dynamics, realistic actuator constraints and environmental disturbances, etc.

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